

Doubly Robust Conditional Independence Testing with Generative Neural Networks

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The “Big Trap”: CI Testing in High Dimensions

Setup. Three random variables X, Y, Z . We ask: *Given the information in Z , is there still any relationship between X and Y ?*

$$H_0 : X \perp\!\!\!\perp Y \mid Z \quad \text{vs.} \quad H_1 : X \not\perp\!\!\!\perp Y \mid Z.$$

An Intuitive Example: Let X = “child’s shoe size”, Y = “child’s vocabulary size”, Z = “age”. Given age Z , shoe size and vocabulary have *no relationship* — they are conditionally independent.

The Traditional Dead End

Classical statistics handles this by estimating $\hat{f}(x, y \mid z)$. But once Z becomes slightly high-dimensional ($d_Z > 5$), methods like **Kernel Density Estimation (KDE)** **completely break down** — you cannot even compute the density reliably, let alone perform a hypothesis test.

The Breakthrough: Use “Fakes” Instead of Densities!

Key insight: Since estimating the density is too hard, *stop trying*. Use neural networks to **generate** the data instead!

The idea (assume H_0 is true):

- ① Train a GNN to generate fake X 's based only on $Z \Rightarrow \tilde{X} = \hat{G}_X(\eta, Z)$
- ② Train another GNN to generate fake Y 's based only on $Z \Rightarrow \tilde{Y} = \hat{G}_Y(\kappa, Z)$

Lemma 1

Given joint distribution P_{XYZ} , there exist random vectors $\tilde{X} \in \mathbb{R}^{d_x}$ and $\tilde{Y} \in \mathbb{R}^{d_y}$ satisfying:

- ① $(\tilde{X}, \tilde{Y}) \perp\!\!\!\perp (X, Y) \mid Z$
- ② $\tilde{X} \perp\!\!\!\perp \tilde{Y} \mid Z$
- ③ $P_{\tilde{X}|Z=z} = P_{X|Z=z}$ and $P_{\tilde{Y}|Z=z} = P_{Y|Z=z}$

Reformulation There exist $\tilde{X}, \tilde{Y}$ such that $P_{\tilde{X}|Z} = P_{X|Z}$, $P_{\tilde{Y}|Z} = P_{Y|Z}$, and $\tilde{X} \perp\!\!\!\perp \tilde{Y} \mid Z$ always. Then:

$$H_0 \iff P_{XYZ} = P_{\tilde{X}\tilde{Y}Z}, \quad H_1 \iff P_{XYZ} \neq P_{\tilde{X}\tilde{Y}Z}.$$

Claim: $X \perp\!\!\!\perp Y \mid Z \iff P_{XYZ} = P_{\tilde{X}\tilde{Y}Z}$.

By Property 2 of Lemma 1 ($\tilde{X} \perp\!\!\!\perp \tilde{Y} \mid Z$) and Property 3 ($P_{\tilde{X}|Z} = P_{X|Z}$, $P_{\tilde{Y}|Z} = P_{Y|Z}$):

$$P_{\tilde{X}\tilde{Y}|Z} \stackrel{\text{P.2}}{=} P_{\tilde{X}|Z} P_{\tilde{Y}|Z} \stackrel{\text{P.3}}{=} P_{X|Z} P_{Y|Z}.$$

This single chain of equalities drives both directions of the proof.

If $H_0: X \perp\!\!\!\perp Y \mid Z$

$$P_{XY|Z} \stackrel{H_0}{=} P_{X|Z} P_{Y|Z} = P_{\tilde{X}\tilde{Y}|Z} \implies \boxed{P_{XYZ} = P_{\tilde{X}\tilde{Y}Z}}$$

If $H_1: X \not\perp\!\!\!\perp Y \mid Z$

$$P_{XY|Z} \stackrel{H_1}{\neq} P_{X|Z} P_{Y|Z} = P_{\tilde{X}\tilde{Y}|Z} \implies \boxed{P_{XYZ} \neq P_{\tilde{X}\tilde{Y}Z}}$$

How to Measure “Looking the Same”?

We need a “**ruler**” to measure the distance between P_{XYZ} and $P_{\tilde{X}\tilde{Y}Z}$. The authors use Maximum Mean Discrepancy (MMD) based CI measure, **MMDCI in short**. Let K_X, K_Y, K_Z be symmetric positive-definite kernels with RKHSs $\mathcal{H}_X, \mathcal{H}_Y, \mathcal{H}_Z$. Define the product kernel $K_0 = K_X \otimes K_Y \otimes K_Z$.

Definition of MMDCI

$$\text{MMDCI}(P_{XYZ}) := \left\| \mathbb{E}[K_0(X, Y, Z, \cdot)] - \mathbb{E}[K_0(\tilde{X}, \tilde{Y}, Z, \cdot)] \right\|_{\mathcal{H}_0}^2$$

Characteristic Kernel:

A kernel K_0 is called a **characteristic kernel** if its corresponding **mean embedding map** $\mu : \mathcal{P} \rightarrow \mathcal{H}_0$ is *injective*, i.e., different distributions are always mapped to different points in the RKHS.

Formally, for any two probability distributions P and Q :

$$\mathbb{E}_P[K_0((X, Y, Z), \cdot)] = \mathbb{E}_Q[K_0((X, Y, Z), \cdot)] \implies P = Q.$$

Equivalently: $\text{MMD}(P, Q) = 0 \iff P = Q$.

How to Measure “Looking the Same”?

Assumption 1: Assumes $K_0 = K_X \otimes K_Y \otimes K_Z$ is a characteristic kernel. **Gaussian** and **Laplacian** kernels are characteristic kernel.

Lemma 2

Under Assumption 1:

$$X \perp\!\!\!\perp Y \mid Z \iff \text{MMDCI}(P_{XYZ}) = 0.$$

$$\text{MMDCI}(P_{XYZ}) := \|\mathbb{E}[K_0(X, Y, Z, \cdot)] - \mathbb{E}[K_0(\tilde{X}, \tilde{Y}, Z, \cdot)]\| \quad (1)$$

$$\begin{aligned} &:= \mathbb{E}[K_0((X, Y, Z), (X', Y', Z'))] - 2\mathbb{E}[K_0((X, Y, Z), (\tilde{X}', \tilde{Y}', Z'))] \\ &\quad + \mathbb{E}[K_0((\tilde{X}, \tilde{Y}, Z), (\tilde{X}', \tilde{Y}', Z'))] \end{aligned} \quad (2)$$

where $(X', Y', Z', \tilde{X}', \tilde{Y}')$ is an independent copy of $(X, Y, Z, \tilde{X}, \tilde{Y})$.

Step 1: Write MMDCI in abstract form. Define mean embeddings:

$$\mu := \mathbb{E}[K_0((X, Y, Z), \cdot)], \quad \nu := \mathbb{E}[K_0((\tilde{X}, \tilde{Y}, Z), \cdot)].$$

Then Eq.(1) becomes: $\text{MMDCI}(P_{XYZ}) = \|\mu - \nu\|_{\mathcal{H}_0}^2$

Step 2: Expand the squared norm. In any Hilbert space:

$$\|\mu - \nu\|_{\mathcal{H}_0}^2 = \langle \mu, \mu \rangle_{\mathcal{H}_0} - 2\langle \mu, \nu \rangle_{\mathcal{H}_0} + \langle \nu, \nu \rangle_{\mathcal{H}_0}.$$

So Eq.(1) naturally expands into **three terms**.

Step 3: Convert inner products to expectations via reproducing property.

The key is the **reproducing property** of RKHS: for any two points w, w' ,

$$\langle K_0(w, \cdot), K_0(w', \cdot) \rangle_{\mathcal{H}_0} = K_0(w, w').$$

Proof From Eq.(1) to Eq.(2)

Applying the reproducing property to each of the three terms:

Term 1: Introduce an i.i.d. copy (X', Y', Z') of (X, Y, Z) :

$$\begin{aligned}\langle \mu, \mu \rangle_{\mathcal{H}_0} &= \langle \mathbb{E}[K_0((X, Y, Z), \cdot)], \mathbb{E}[K_0((X', Y', Z'), \cdot)] \rangle_{\mathcal{H}_0} \\ &= \mathbb{E}[K_0((X, Y, Z), (X', Y', Z'))].\end{aligned}$$

Term 2: Introduce an i.i.d. copy $(\tilde{X}', \tilde{Y}', Z')$ of $(\tilde{X}, \tilde{Y}, Z)$:

$$\langle \mu, \nu \rangle_{\mathcal{H}_0} = \mathbb{E}[K_0((X, Y, Z), (\tilde{X}', \tilde{Y}', Z'))].$$

Term 3:

$$\langle \nu, \nu \rangle_{\mathcal{H}_0} = \mathbb{E}[K_0((\tilde{X}, \tilde{Y}, Z), (\tilde{X}', \tilde{Y}', Z'))].$$

Combining all three terms yields **Eq.(2)**:

$$\begin{aligned}\text{MMDCI}(P_{XYZ}) &= \mathbb{E}[K_0((X, Y, Z), (X', Y', Z'))] \\ &\quad - 2 \mathbb{E}[K_0((X, Y, Z), (\tilde{X}', \tilde{Y}', Z'))] \\ &\quad + \mathbb{E}[K_0((\tilde{X}, \tilde{Y}, Z), (\tilde{X}', \tilde{Y}', Z'))].\end{aligned}$$

The Problem with Eq.(2)

Eq.(2) still contains $(\tilde{X}, \tilde{Y})$ and their independent copies $(\tilde{X}', \tilde{Y}')$. These are **unobservable theoretical variables** — they exist only to build the null hypothesis structure, and **cannot be computed from data** $\{(X_i, Y_i, Z_i)\}$.

Goal: *Eliminate the tilde variables* by replacing them with quantities that only depend on X, Y, Z and their conditional distributions.

Key tool: Iterated Expectation + Product Kernel Factorization

Define the **conditional kernel mean embeddings**:

$$g_X(z) := \mathbb{E}[K_X(X, \cdot) \mid Z = z] \in \mathcal{H}_X, \quad g_Y(z) := \mathbb{E}[K_Y(Y, \cdot) \mid Z = z] \in \mathcal{H}_Y.$$

These are the *conditional mean embeddings* of $X \mid Z$ and $Y \mid Z$ into their RKHSs — computable quantities that fully characterize $P_{X|Z}$ and $P_{Y|Z}$.

Recall Eq.(2):

$$\begin{aligned} \text{MMDCI}(P_{XYZ}) = & \underbrace{\mathbb{E}[K_0((X, Y, Z), (X', Y', Z'))]}_{\text{Term I}} - 2 \underbrace{\mathbb{E}[K_0((X, Y, Z), (\tilde{X}', \tilde{Y}', Z'))]}_{\text{Term II}} \\ & + \underbrace{\mathbb{E}[K_0((\tilde{X}, \tilde{Y}, Z), (\tilde{X}', \tilde{Y}', Z'))]}_{\text{Term III}} \end{aligned}$$

Key tools used throughout:

- **Product factorization:** $K_0 = K_X \otimes K_Y \otimes K_Z$, i.e.
 $K_0((x, y, z), (x', y', z')) = K_X(x, x') K_Y(y, y') K_Z(z, z')$
- **Reproducing property:** $\langle K_X(x, \cdot), K_X(x', \cdot) \rangle_{\mathcal{H}_X} = K_X(x, x')$
- **Definition:** $g_X(z) := \mathbb{E}[K_X(X, \cdot) \mid Z = z] \in \mathcal{H}_X$

Goal

Eliminate all tilde variables $\tilde{X}, \tilde{Y}, \tilde{X}', \tilde{Y}'$ and express everything in terms of X, Y, Z, X', Y', Z' and g_X, g_Y only.

Term I contains no tilde variables, so it stays as is. Using the product factorization $K_0 = K_X \otimes K_Y \otimes K_Z$:

$$\mathbb{E}[K_0((X, Y, Z), (X', Y', Z'))] = \mathbb{E}[K_X(X, X') K_Y(Y, Y') K_Z(Z, Z')].$$

This is **directly computable** from the observed data — no tilde variables involved.

Term II: Eliminate $\tilde{X}', \tilde{Y}'$

Term II: $\mathbb{E}[K_0((X, Y, Z), (\tilde{X}', \tilde{Y}', Z'))]$

Apply product factorization first: $\mathbb{E}[K_X(X, \tilde{X}') K_Y(Y, \tilde{Y}') K_Z(Z, Z')]$ Since $(\tilde{X}', \tilde{Y}') \perp\!\!\!\perp (X, Y) \mid Z'$, and $\tilde{X}' \perp\!\!\!\perp \tilde{Y}' \mid Z'$, take $\mathbb{E}[\cdot \mid X, Y, Z, Z']$:

$$= \mathbb{E}\left[\mathbb{E}[K_X(X, \tilde{X}') \mid Z'] \cdot \mathbb{E}[K_Y(Y, \tilde{Y}') \mid Z'] \cdot K_Z(Z, Z')\right].$$

Now expand the key step for $\mathbb{E}[K_X(X, \tilde{X}') \mid X, Z']$:

$$\begin{aligned}\mathbb{E}[K_X(X, \tilde{X}') \mid X, Z'] &= \mathbb{E}[K_X(X, \tilde{X}') \mid Z']|_{x=X} \quad (\text{condition on } Z' \text{ only, since } \tilde{X}' \perp\!\!\!\perp X \mid Z') \\ &= \mathbb{E}[K_X(x, \tilde{X}') \mid Z']|_{x=X} \quad (\tilde{X}' \mid Z' \sim X \mid Z') \\ &= \mathbb{E}[\langle K_X(X, \cdot), K_X(x, \cdot) \rangle_{\mathcal{H}_X} \mid Z']|_{x=X} \quad (\text{reproducing property}) \\ &= \langle \mathbb{E}[K_X(X, \cdot) \mid Z'], K_X(x, \cdot) \rangle_{\mathcal{H}_X}|_{x=X} \quad (\text{linearity of } \langle \cdot, \cdot \rangle) \\ &= \langle g_X(Z'), K_X(X, \cdot) \rangle_{\mathcal{H}_X}.\end{aligned}$$

By the same argument for Y : $\mathbb{E}[K_Y(Y, \tilde{Y}') \mid Y, Z'] = \langle g_Y(Z'), K_Y(Y, \cdot) \rangle_{\mathcal{H}_Y}$.

Therefore:

$$\text{Term II} = \mathbb{E}[\langle g_X(Z'), K_X(X, \cdot) \rangle_{\mathcal{H}_X} \langle g_Y(Z'), K_Y(Y, \cdot) \rangle_{\mathcal{H}_Y} K_Z(Z, Z')].$$

Term III: Eliminate $\tilde{X}, \tilde{Y}, \tilde{X}', \tilde{Y}'$

Term III: $\mathbb{E}[K_0((\tilde{X}, \tilde{Y}, Z), (\tilde{X}', \tilde{Y}', Z'))]$

Apply product factorization:

$$= \mathbb{E}[K_X(\tilde{X}, \tilde{X}') K_Y(\tilde{Y}, \tilde{Y}') K_Z(Z, Z')].$$

Condition on Z, Z' , then use $\tilde{X} \perp\!\!\!\perp \tilde{Y} \mid Z$ and $\tilde{X}' \perp\!\!\!\perp \tilde{Y}' \mid Z'$:

$$= \mathbb{E}\left[\mathbb{E}[K_X(\tilde{X}, \tilde{X}') \mid Z, Z'] \cdot \mathbb{E}[K_Y(\tilde{Y}, \tilde{Y}') \mid Z, Z'] \cdot K_Z(Z, Z')\right].$$

Now apply the same derivation chain to $\mathbb{E}[K_X(\tilde{X}, \tilde{X}') \mid Z, Z']$:

$$\begin{aligned}\mathbb{E}[K_X(\tilde{X}, \tilde{X}') \mid Z, Z'] &= \mathbb{E}[\langle K_X(\tilde{X}, \cdot), K_X(\tilde{X}', \cdot) \rangle_{\mathcal{H}_X} \mid Z, Z'] \quad (\text{reproducing property}) \\ &= \langle \mathbb{E}[K_X(\tilde{X}, \cdot) \mid Z], \mathbb{E}[K_X(\tilde{X}', \cdot) \mid Z'] \rangle_{\mathcal{H}_X} \quad (\tilde{X} \perp\!\!\!\perp \tilde{X}' \mid Z, Z') \\ &= \langle \mathbb{E}[K_X(X, \cdot) \mid Z], \mathbb{E}[K_X(X, \cdot) \mid Z'] \rangle_{\mathcal{H}_X} \\ &= \langle g_X(Z), g_X(Z') \rangle_{\mathcal{H}_X}.\end{aligned}$$

By the same argument: $\mathbb{E}[K_Y(\tilde{Y}, \tilde{Y}') \mid Z, Z'] = \langle g_Y(Z), g_Y(Z') \rangle_{\mathcal{H}_Y}$.

Therefore: Term III = $\mathbb{E}[\langle g_X(Z), g_X(Z') \rangle_{\mathcal{H}_X} \langle g_Y(Z), g_Y(Z') \rangle_{\mathcal{H}_Y} K_Z(Z, Z')]$

Combining All Three Terms: Eq.(3)

Substituting Terms I, II, III back into Eq.(2):

Eq.(3) — MMDCI in terms of g_X, g_Y only

$$\begin{aligned} \text{MMDCI}(P_{XYZ}) = \mathbb{E} \Big[& \left(K_X(X, X') \cdot K_Y(Y, Y') \right. \\ & - \langle g_X(Z'), K_X(X, \cdot) \rangle_{\mathcal{H}_X} \cdot \langle g_Y(Z'), K_Y(Y, \cdot) \rangle_{\mathcal{H}_Y} \\ & - \langle g_X(Z), K_X(X', \cdot) \rangle_{\mathcal{H}_X} \cdot \langle g_Y(Z), K_Y(Y', \cdot) \rangle_{\mathcal{H}_Y} \\ & \left. + \langle g_X(Z), g_X(Z') \rangle_{\mathcal{H}_X} \cdot \langle g_Y(Z), g_Y(Z') \rangle_{\mathcal{H}_Y} \right) \cdot K_Z(Z, Z') \Big], \end{aligned}$$

Eq.(2) — not computable Contains $\tilde{X}, \tilde{Y}, \tilde{X}', \tilde{Y}'$ (unobservable oracle variables)

Eq.(3) — computable! Only involves X, Y, Z, X', Y', Z' and $g_X(\cdot), g_Y(\cdot)$ (estimable by GNNs)

Next step: Replace g_X, g_Y with GNN estimators $\hat{g}_X, \hat{g}_Y \Rightarrow$ construct the sample U-statistic $\hat{T}_J$.

Eq.(4): Doubly Robust Form

$$\text{MMDCI}(P_{XYZ}) = \mathbb{E}[U(X, X') V(Y, Y') K_Z(Z, Z')], \quad \text{with}$$

$$U(X, X') = K_X(X, X') - \langle g_X(Z), K_X(X', \cdot) \rangle_{\mathcal{H}_X} - \langle g_X(Z'), K_X(X, \cdot) \rangle_{\mathcal{H}_X} \\ + \langle g_X(Z), g_X(Z') \rangle_{\mathcal{H}_X},$$

$$V(Y, Y') = K_Y(Y, Y') - \langle g_Y(Z), K_Y(Y', \cdot) \rangle_{\mathcal{H}_Y} - \langle g_Y(Z'), K_Y(Y, \cdot) \rangle_{\mathcal{H}_Y} \\ + \langle g_Y(Z), g_Y(Z') \rangle_{\mathcal{H}_Y}.$$

Intuition for U and V (think of them as *kernel-space regression residuals*):

- U = similarity between true observed X 's, *minus* the similarity we “predict” X would have given Z .
- V = similarity between true observed Y 's, *minus* the similarity we “predict” Y would have given Z .

Property 1: Multiplicative Structure

$$\text{MMDCI}(P_{XYZ}) = \mathbb{E}[U(X, X') \cdot V(Y, Y') \cdot K_Z(Z, Z')]$$

U depends only on X, X', Z, Z' via g_X ; V depends only on Y, Y', Z, Z' via g_Y .

The two nuisance functions g_X and g_Y enter *separately and symmetrically*.

Property 2: U and V are Zero-Mean and Symmetric

$$\mathbb{E}[U(X, X') \mid Z, Z'] = 0, \quad \mathbb{E}[V(Y, Y') \mid Z, Z'] = 0.$$

Both U and V are symmetric in their arguments: $U(X, X') = U(X', X)$ and $V(Y, Y') = V(Y', Y)$.

Property 3: $U \perp\!\!\!\perp V \mid (Z, Z')$ under H_0 Under H_0 : $X \perp\!\!\!\perp Y \mid Z$, so U and V are conditionally independent given (Z, Z') . Therefore:

$$\mathbb{E}[U(X, X') \cdot V(Y, Y') \mid Z, Z'] \stackrel{H_0}{=} \mathbb{E}[U \mid Z, Z'] \cdot \mathbb{E}[V \mid Z, Z'] = 0 \cdot 0 = 0,$$

which implies $\text{MMDCI}(P_{XYZ}) = 0$ under H_0 .



Definition 1 (Double Robustness for CI Testing)

A population-level measure $D(P_{XYZ})$ is called **doubly robust** against misspecifications of $g_X = \mathbb{E}[K_X(X, \cdot) | Z]$ and $g_Y = \mathbb{E}[K_Y(Y, \cdot) | Z]$ if, under H_0 :

$$D(P_{XYZ}) = 0 \quad \text{whenever } \hat{g}_X = g_X \text{ or } \hat{g}_Y = g_Y.$$

That is, the measure remains valid as long as *at least one* of the two conditional mean embeddings is correctly specified.

Why Eq.(4) satisfies Definition 1

Suppose $\hat{g}_X \neq g_X$ (large error) but $\hat{g}_Y = g_Y$ (exact). Then under H_0 : $\mathbb{E}[V | Z, Z'] = 0$, so

$$\mathbb{E}[U_{\hat{g}_X}(X, X') \cdot V_{g_Y}(Y, Y') \cdot K_Z(Z, Z')] = \mathbb{E}[U_{\hat{g}_X} \cdot \underbrace{\mathbb{E}[V_{g_Y} | Z, Z']}_{=0} \cdot K_Z] = 0.$$

Error $\times 0 = 0$. The reverse holds by symmetry.

From the reproducing property of the RKHS:

$$\langle g_X(Z), K_X(X, \cdot) \rangle_{\mathcal{H}_X} = \mathbb{E}_{X'}[K_X(X, X') \mid Z], \quad \langle g_X(Z), g_X(Z') \rangle_{\mathcal{H}_X} = \mathbb{E}_{X, X'}[K_X(X, X') \mid Z, Z'].$$

Eq.(5): Oracle statistic T^*

$$T^* = \frac{1}{n(n-1)} \sum_{k \neq \ell} U^*(X_k, X_\ell) V^*(Y_k, Y_\ell) K_Z(Z_k, Z_\ell), \quad \text{with}$$

$$U^*(X_k, X_\ell) = K_X(X_k, X_\ell) - \mathbb{E}_X[K_X(X, X_k) \mid Z = Z_\ell] - \mathbb{E}_X[K_X(X, X_\ell) \mid Z = Z_k] \\ + \mathbb{E}_{X, X'}[K_X(X, X') \mid Z = Z_k, Z' = Z_\ell],$$

$$V^*(Y_k, Y_\ell) = K_Y(Y_k, Y_\ell) - \mathbb{E}_Y[K_Y(Y, Y_k) \mid Z = Z_\ell] - \mathbb{E}_Y[K_Y(Y, Y_\ell) \mid Z = Z_k] \\ + \mathbb{E}_{Y, Y'}[K_Y(Y, Y') \mid Z = Z_k, Z' = Z_\ell].$$

T^* is a **degenerate U-statistic of order 2** under H_0 , since $\mathbb{E}[U(X_k, X_\ell) \mid Z_k, Z_\ell] = 0$ under H_0 .

1. U-Statistic: A U-statistic plugs all *non-repeated pairs* from the sample into a kernel $h(\cdot, \cdot)$ and averages. For the order-2 case:

$$U_n = \frac{1}{n(n-1)} \sum_{i \neq j} h(W_i, W_j).$$

2. Order 2: each term uses exactly two observations (X_k, X_ℓ) , (Y_k, Y_ℓ) , (Z_k, Z_ℓ) .

3. Degenerate: First-Order Conditional Mean is Zero Under H_0

Under H_0 , the first-order projection of the kernel vanishes:

$$\mathbb{E}[U^*(X_k, X_\ell) V^*(Y_k, Y_\ell) K_Z(Z_k, Z_\ell) \mid X_k, Y_k, Z_k] = 0.$$

Problem: U^* and V^* require intractable conditional expectations $\mathbb{E}[K_X(X, \cdot) \mid Z]$. We replace them with a trained GNN generator $\hat{\mathbb{G}}_X$ that draws Monte Carlo samples from $P_{X|Z}$.

Step 1: Train the GNN on the Training Fold $[n_T]$

Find the generator that minimises the empirical U-statistic of $\hat{U}$ over the training set (cross-fitting fold of size n_T), given M i.i.d. with $\eta_k^{(m)} \sim \mathcal{N}(0, I_m)$:

$$\hat{\mathbb{G}}_X = \arg \min_{\mathbb{G}_X \in \mathcal{G}_X} \frac{1}{n_T(n_T - 1)} \sum_{\substack{k \neq \ell \\ k, \ell \in [n_T]}} \hat{U}(X_k, X_\ell) \cdot K_Z(Z_k, Z_\ell).$$

$$\begin{aligned} \hat{U}(X_k, X_\ell) &= K_X(X_k, X_\ell) - \frac{1}{M} \sum_{m=1}^M K_X(X_k, \mathbb{G}_X(\eta_\ell^m, Z_\ell)) - \frac{1}{M} \sum_{m=1}^M K_X(X_\ell, \mathbb{G}_X(\eta_k^m, Z_k)) \\ &\quad + \frac{1}{M^2} \sum_{m_1, m_2=1}^M K_X(\mathbb{G}_X(\eta_k^{m_1}, Z_k), \mathbb{G}_X(\eta_\ell^{m_2}, Z_\ell)). \end{aligned}$$

Step 2: Cross-fitting (J folds): train on $J - 1$ folds, test on the remaining fold, then average over all J roles. Divide $\{1, 2, \dots, n\}$ into J equal folds $\mathcal{J}^{(1)}, \dots, \mathcal{J}^{(J)}$. For fold j , train $\hat{G}_X^{(j)}, \hat{G}_Y^{(j)}$ on $\mathcal{D}_{-j} := \{(X_i, Y_i, Z_i)\}_{i \in \mathcal{J}^{(-j)}}$, then evaluate test statistic on $\mathcal{J}^{(j)}$.

Eq.(6) Final test statistic $\hat{T}_J$

$$\hat{T}_J = \frac{1}{J} \sum_{j=1}^J \left\{ \frac{1}{\frac{n}{J} \left(\frac{n}{J} - 1 \right)} \sum_{\substack{k \neq \ell \\ k, \ell \in \mathcal{J}^{(j)}}} \hat{U}(X_k, X_\ell) \hat{V}(Y_k, Y_\ell) \mathcal{K}_Z(Z_k, Z_\ell) \right\},$$

$$\hat{U}(X_k, X_\ell) = \mathcal{K}_X(X_k, X_\ell) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_X(X_k, \hat{X}_\ell^{(m)}) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_X(X_\ell, \hat{X}_k^{(m)}) + \frac{1}{M^2} \sum_{m_1, m_2=1}^M \mathcal{K}_X(\hat{X}_k^{(m_1)}, \hat{X}_\ell^{(m_2)})$$

$$\hat{V}(Y_k, Y_\ell) = \mathcal{K}_Y(Y_k, Y_\ell) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_Y(Y_k, \hat{Y}_\ell^{(m)}) - \frac{1}{M} \sum_{m=1}^M \mathcal{K}_Y(Y_\ell, \hat{Y}_k^{(m)}) + \frac{1}{M^2} \sum_{m_1, m_2=1}^M \mathcal{K}_Y(\hat{Y}_k^{(m_1)}, \hat{Y}_\ell^{(m_2)}).$$

Why Wild Bootstrap?

The limiting null distribution of $n\hat{T}_J$ is not pivotal — it depends on unknown eigenvalues of kernel operators and the unknown P_{XYZ} . $\Rightarrow$ Use wild bootstrap to approximate the null distribution of $n\hat{T}_J$.

Eq.(7) Bootstrap Construction:

For $b = 1, 2, \dots, B$, draw n i.i.d. multipliers $\{e_{bi}\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$. Define the b -th bootstrap statistic:

$$\hat{T}_J^b = \frac{1}{J} \sum_{j=1}^J \left\{ \frac{1}{\frac{n}{J} \left(\frac{n}{J} - 1 \right)} \sum_{\substack{k \neq \ell \\ k, \ell \in \mathcal{J}^{(j)}}} \hat{U}(X_k, X_\ell) \hat{V}(Y_k, Y_\ell) \mathcal{K}_Z(Z_k, Z_\ell) e_{bk} e_{b\ell} \right\}.$$

The only difference from $\hat{T}_J$: multiply each term by $e_{bk} e_{b\ell}$.

Rejection Rule at Level $\gamma \in (0, 1)$: Reject H_0 if the bootstrap proportion exceeding $\hat{T}_J$ falls below γ :

$$\text{Reject } H_0 \iff \frac{1}{B} \sum_{b=1}^B \mathbf{1}_{\{\hat{T}_J^b > \hat{T}_J\}} < \gamma.$$

The Whole CI Test Procedure

Step 1: Data Generating Process (DGP)

Specify joint distribution of (X, Y, Z) under H_0 or H_1 . Draw n i.i.d. copies $\{(X_i, Y_i, Z_i)\}_{i=1}^n$

Step 2: Cross-Fitting Split

Divide $\{1, \dots, n\}$ into J equal folds $\mathcal{J}^{(1)}, \dots, \mathcal{J}^{(J)}$, For fold j : training set $\mathcal{D}_{-j}$, test set $\mathcal{J}^{(j)}$

Step 3: Train GNN Generators on Each Training Fold

Minimize loss function on $\mathcal{D}_{-j}$ to obtain $\hat{\mathbb{G}}_X^j$ and $\hat{\mathbb{G}}_Y^j$

Generate M synthetic samples per observation: $\hat{X}_i^{(m)} = \hat{\mathbb{G}}_X^j(\eta_i^m, Z_i)$, $\hat{Y}_i^{(m)} = \hat{\mathbb{G}}_Y^j(\kappa_i^m, Z_i)$

Step 4: Compute Test Statistic $\hat{T}_J$

Plug $\hat{X}_i^{(m)}, \hat{Y}_i^{(m)}$ into $\hat{U}, \hat{V}$ on test fold $\mathcal{J}^{(j)}$ and average over J folds to obtain $\hat{T}_J$

Step 5: Wild Bootstrap Calibration

For $b = 1, \dots, B$: draw $\{e_{bi}\} \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$, compute $\hat{T}_J^b$ and reject H_0 at level γ if

$$B^{-1} \sum_{b=1}^B \mathbf{1}_{\{\hat{T}_J^b > \hat{T}_J\}} < \gamma$$

Step 6: Evaluate Size & Power

Size: repeat under H_0 , check empirical rejection rate $\approx \gamma$

Power: repeat under H_1 , measure empirical rejection rate

Assumption 2

Assume $M \rightarrow \infty$ as $n \rightarrow \infty$. There exist constants $C_0 > 0$, $\alpha_1, \alpha_2 > 0$ satisfying $\alpha_1 + \alpha_2 > \frac{1}{2}$, such that for $D_i \in \{X_i, \hat{X}_i^{(1)}, \tilde{X}_i^{(1)}\}$, $E_i \in \{Y_i, \hat{Y}_i^{(1)}, \tilde{Y}_i^{(1)}\}$ and $i \in \{i_1, \dots, i_\ell\}$ where $i_s \in \mathcal{J}^{(s)}$ for each $s \in [J]$, we have:

(a) $\mathbb{E}[\mathcal{K}_X(D_i, D_i) \mathcal{K}_Y(E_i, E_i) \mathcal{K}_Z(Z_i, Z_i)] < C_0.$

(b)

$$\sqrt{\mathbb{E}\left[\left\|\mathbb{E}[\mathcal{K}_X(\cdot, X_i) \mid Z_i] - \mathbb{E}[\mathcal{K}_X(\cdot, \hat{X}_i^{(1)}) \mid Z_i]\right\|_{\mathcal{H}_X}^2 \cdot (\sqrt{\mathcal{K}_Z(Z_i, Z_i)} + \mathcal{K}_Y(E_i, E_i) \mathcal{K}_Z(Z_i, Z_i))\right]} = O(n^{-\alpha_1}),$$

and

$$\sqrt{\mathbb{E}\left[\left\|\mathbb{E}[\mathcal{K}_Y(\cdot, Y_i) \mid Z_i] - \mathbb{E}[\mathcal{K}_Y(\cdot, \hat{Y}_i^{(1)}) \mid Z_i]\right\|_{\mathcal{H}_Y}^2 \cdot (\sqrt{\mathcal{K}_Z(Z_i, Z_i)} + \mathcal{K}_X(D_i, D_i) \mathcal{K}_Z(Z_i, Z_i))\right]} = O(n^{-\alpha_2}).$$

Theorem (1)

Suppose Assumption 2 holds. Under H_0 , as $n, M \rightarrow \infty$,

$$\hat{T}_J = T_J^* + O_p\left(n^{-1}M^{-1/2} + n^{-1}(n^{-\alpha_1} + n^{-\alpha_2}) + n^{-1/2-(\alpha_1+\alpha_2)}\right)$$

and

$$n\hat{T}_J \xrightarrow{\mathcal{D}} T^\dagger = \sum_{j=1}^J T_j^\dagger, \quad \text{where} \quad T_1^\dagger \stackrel{d}{=} \sum_{s=1}^{\infty} \lambda_s (\chi_s^2 - 1).$$

- $T_j^\dagger$ are i.i.d., each a weighted sum of centred χ_1^2 random variables.
- $\{\lambda_s\}_{s \geq 1}$ are eigenvalues of the compact self-adjoint operator on $L^2(\mathbb{R}^{d_X+d_Y+d_Z}, P_{XYZ})$ induced by the kernel

$$h((X_1, Y_1, Z_1), (X_2, Y_2, Z_2)) = U(X_1, X_2) V(Y_1, Y_2) K_Z(Z_1, Z_2).$$

- The key condition $\alpha_1 + \alpha_2 > \frac{1}{2}$ (Assumption 2) ensures the approximation error is $o_p(n^{-1})$, so the plug-in statistic inherits the oracle null distribution.

Motivation: Why is Pure MMD Training Insufficient?

- ▶ Original GNN trains solely via MMD loss — a “macro-level” distributional constraint

Our Modification: Add a Mean Alignment Penalty

- ▶ Force the generator's first conditional moment to be accurate:

Original objective: $\min \mathcal{L}_{\text{MMD}}(P_{\text{true}}, P_{\text{fake}})$

Modified objective: $\min \mathcal{L}_{\text{MMD}} + \lambda \cdot \text{MSE}(\mathbb{E}[\hat{X} | Z], X)$

↪ MSE term approximated via Monte Carlo: generate M_0 fake samples per Z_i , average them as

$$\hat{\mathbb{E}}[X | Z_i] = \frac{1}{M_0} \sum_{m=1}^{M_0} \hat{X}_i^{(m)}$$

More possible alignment mechanism: median alignment / multiple quantile alignment ...

Thank You

Questions welcome.